Model Implementation and Testing

Chapter 5 - CAPM, APT, and Diagnostic Testing in Practice

[Model Implementation and Testing | Springer Nature Link]

Chapter 5 is BETTER DEMONSTRATED than presented as PPT

However, we will cover some of the finer points in ppt.

Objectives

- Implement CAPM and APT models in Python
- Understand rolling beta and adjusted beta calculations
- Master hypothesis testing in regression models
- Learn diagnostic tests for model validation
- Implement quantile regression
- Apply Fama-French factor models
- Understand the January Effect

Optimal Hedge Ratio

Definition:

- Percentage of hedging instrument to use
- Minimizes portfolio variance
- Complements CAPM framework

Formula:

$$\text{OHR} = \rho \times \frac{\sigma_s}{\sigma_f}$$

Where:

- ρ = Correlation between spot and futures price changes
- σ_s = Standard deviation of spot price changes
- σ_f = Standard deviation of futures price changes

Regression Form:

$$R_{stock} = \alpha + \beta R_{futures} + \epsilon$$

Beta is the hedge ratio!

Rolling Beta

Purpose:

- Dynamic view of beta over time
- Track changing risk profile

Advantages:

- Captures changing market relationships
- Identifies periods of high/low sensitivity
- Useful for risk management

Adjusted Beta

Purpose: Account for mean reversion (betas tend toward 1 over time)

Formula:

$$\beta_{adjusted} = \frac{1}{3} \times \beta_{raw} + \frac{2}{3} \times 1$$

$$\beta_{adjusted} = 0.67 \times \beta_{raw} + 0.33 \times 1$$

Example:

- If raw β = 1.50
- Adjusted β = 0.67 × 1.50 + 0.33 = 1.34

Interpretation: Less extreme than raw beta

Use: Better estimate of future beta

Scholes-Williams Beta

Purpose:

- Correct for non-synchronous trading
- Uses lagged and lead market returns

Steps:

Contemporaneous Beta: $\beta_0 = \frac{cov(R_i,R_m)}{\sigma(R_m)}$

Lagged Beta: $\beta_{-1} = \frac{cov(R_i,R_{m,-1})}{\sigma(R_m)}$

Lead Beta: $\beta_{+1} = \frac{cov(R_i,R_{m,+1})}{\sigma(R_m)}$

SW Beta: $\beta_{SW} = \frac{\beta_0+\beta_{-1}+\beta_{+1}}{3}$

Comparison:

Type	Value	Interpretation
Contemporaneous	1.130	Standard beta
Lagged	-0.046	Weak inverse relationship
Lead	-0.046	Weak inverse relationship
SW Adjusted	0.346	Corrected for timing issues

Quantile Regression

OLS vs Quantile Regression

Aspect	**OLS**	**Quantile Regression**
Focus	Conditional mean	Conditional quantiles
Goal	Minimize squared errors	Minimize weighted absolute errors
Distribution	Assumes normality	No distributional assumptions
Robustness	Sensitive to outliers	Robust to outliers
Insight	Average relationship	Relationship across distribution

Key Insight:

- Understand relationships outside the mean
- Useful for non-normal distributions
- Capture different effects at different quantiles

Key takeaways

1. CAPM Implementation
- Beta estimation via regression
- Rolling beta for time-varying risk
- Adjusted beta for mean reversion

2. Hypothesis Testing
- T-tests for individual coefficients
- F-tests for joint hypotheses
- p-value interpretation critical

3. Quantile Regression
- Understand relationships across distribution
- Robust to outliers
- No normality assumptions

4. Fama-French Models
- Multiple factors improve explanation
- SMB (size) and HML (value) factors
- Data from Ken French Library

5. APT Implementation
- Multiple macroeconomic factors
- Significant: Market, Inflation, IP, Term Spread
- Non-significant: Yield Spread, Consumer Credit

6. Diagnostic Testing
- Always test assumptions
- Non-normality, heteroscedasticity, autocorrelation
- Use robust standard errors when needed

7. Model Refinement
- Influential point detection (Cook's Distance)
- Dummy variables for outliers
- Polynomial/interaction terms

8. Stability Checks
- CUSUM, Chow, rolling regression
- Parameter stability important
- Structural breaks can invalidate model

9. Seasonal Effects
- January effect exists in energy markets
- Significant coefficient (0.050, $p = 0.039$)
- Improves model fit

Questions & Discussion

Review Questions:

- What is the difference between rolling beta and adjusted beta?
- When would you use quantile regression instead of OLS?
- What do the Fama-French factors (SMB and HML) represent?
- How do you test for heteroscedasticity?
- What is the January effect and how is it tested?

Discussion Points:

- Choosing between CAPM, Fama-French, and APT
- Balancing model complexity and interpretability
- Handling influential observations
- Importance of diagnostic testing